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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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Experiment – 10: Orchestration using Kubernetes

Aim: To study and implement container orchestration using Kubernetes on AWS/ Azure/ Google cloud platform (Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to learn –
- fundamental concepts of containerization technologies like kubernetes
- how to provision and manage container on AWS cloud with EKS Elastic Kubernetes Services.
- how to use eksctl, docker and kubectl command line
- how to create and use EKS Cluster

Lab objective mapped: ITDO8024.6 : To design and develop containerized solutions.

Prerequisite: An AWS account

Requirements:
- Must have Hyper V capabilities on the system
- Minikube installed
- Idea of system administration and linux commands
- Understands the basic concepts of containerization technology
- Cloud Login, Desktop, Browser, Internet

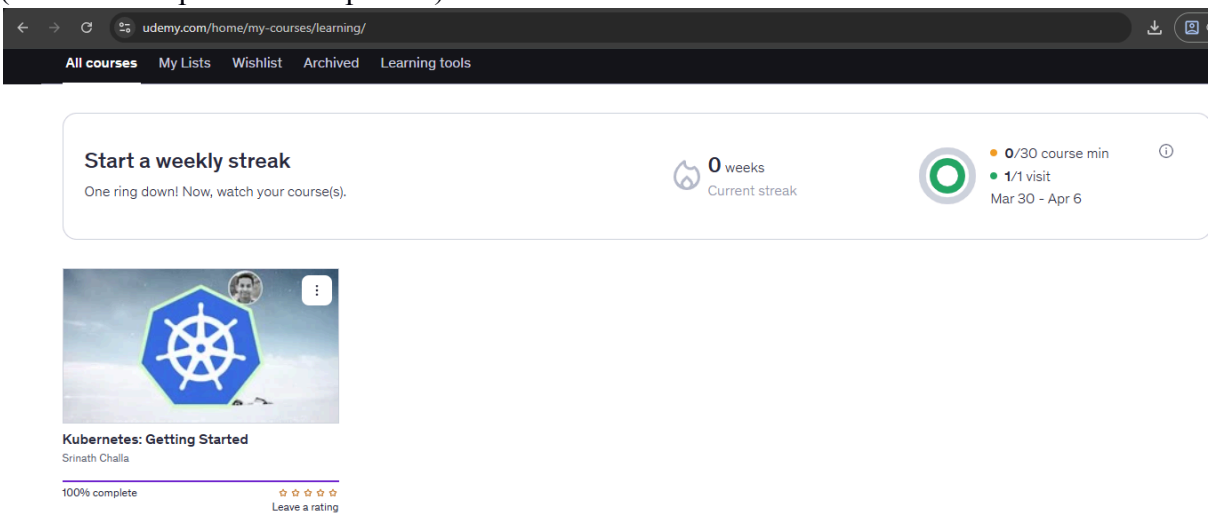
Pre-Experiment Theory: (To be written and drawn in hand)
1. **Amazon EKS**
2. **Architecture of Kubernetes**

Procedure:

Follow given aws documentation link to complete Expt 10
https://docs.aws.amazon.com/eks/latest/userguide/getting-started-console.html

Post-Experiments Exercise:
Complete Certification Course:

I. https://www.udemy.com/course/kubernetes-getting-started/
(Attach some proof of completion)



Extended Theory: (to be written in hand)
Kubernetes with AWS Elastic Kubernetes Services (EKS) (Udemy Video):

I. https://www.udemy.com/course/master-kubernetes-with-aws-elastic-kubernetes-services-eks/
Watch Preview videos
1. What is Kubernetes and EKS
2. Setting up environment for Kubernetes

Results/Calculations/Observations:
Fill the following observation tables-(to be written in hand)

Sr. No	**Item**	**Answer**
1.	For 2 Amazon EKS cluster that you create in Asia Pacific (Mumbai) how much does that charge	Amazon EKS charges about $0.10 per cluster per hour in the Asia Pacific (Mumbai) region. For 2 clusters, the cost is approximately $0.20 per hour, which comes to around $144 per month (0.20 × 24 × 30). This cost is only for the control plane, and additional charges apply for resources like EC2 instances, storage, and data transfer.

2.	kubectl- Which tool is this? its use	kubectl is a command-line tool used to interact with Kubernetes clusters. It allows users to deploy applications, view and manage cluster resources, check logs, and control workloads by sending commands to the Kubernetes API server.

Conclusion:
1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:
[1] [Online]https://www.bluematador.com
[2] [Online] https://docs.aws.amazon.com/eks/latest/userguide/what-is-eks.html
[3] [Online] https://www.eksworkshop.com/docs/introduction
[4] [Online] https://aws.amazon.com/getting-started/hands-on/deploy-kubernetes-app-amazon-eks/

```
C:\Users\Student>aws sts get-caller-identity
{
    "UserId": "AIDAEXAMPLE123456",
    "Account": "987654321000",
    "Arn": "arn:aws:iam::987654321000:user/student-user"
}
```

```
C:\Users\Student>aws cloudformation create-stack --region ap-south-1 --stack-name my-eks-vp
{
    "StackId": "arn:aws:cloudformation:ap-south-1:987654321000:stack/my-eks-vpc-stack/abcd1
}
```

```
C:\Users\Student>aws cloudformation describe-stacks --stack-name my-eks-vpc-stack
{
    "Stacks": [
        {
            "StackName": "my-eks-vpc-stack",
            "CreationTime": "2026-04-01T10:15:30.000Z",
            "StackStatus": "CREATE_COMPLETE"
        }
    ]
}
```